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Chen et al.

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(54) **STRUCTURE AND METHOD RELATED TO A POWER MODULE USING A HYBRID SPACER**

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H01L 21/56 (2006.01)
H01L 23/00 (2006.01)
H01L 23/31 (2006.01)
H01L 23/373 (2006.01)
H05K 7/20 (2006.01)

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CPC **H01L 23/3675** (2013.01); **H01L 21/565** (2013.01); **H01L 23/3107** (2013.01); **H01L 23/3735** (2013.01); **H01L 24/32** (2013.01); **H05K 7/2089** (2013.01); **H01L 2224/32245** (2013.01); **H01L 2224/84801** (2013.01); **H01L 2924/13055** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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Primary Examiner — Sue A Purvis

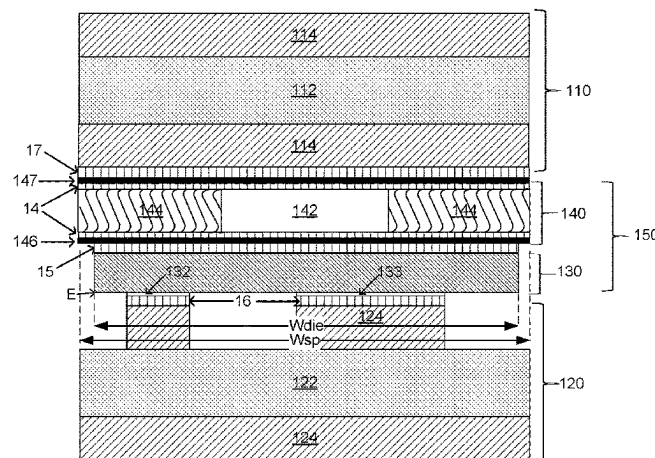
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(57) **ABSTRACT**

In at least one aspect, a method can include shaping a block of flexible spacer material. The method can include shaping a portion of the block of flexible spacer material to receive a solid metal block. The method can include coupling the solid metal block to the portion of the block of flexible spacer material.

20 Claims, 17 Drawing Sheets



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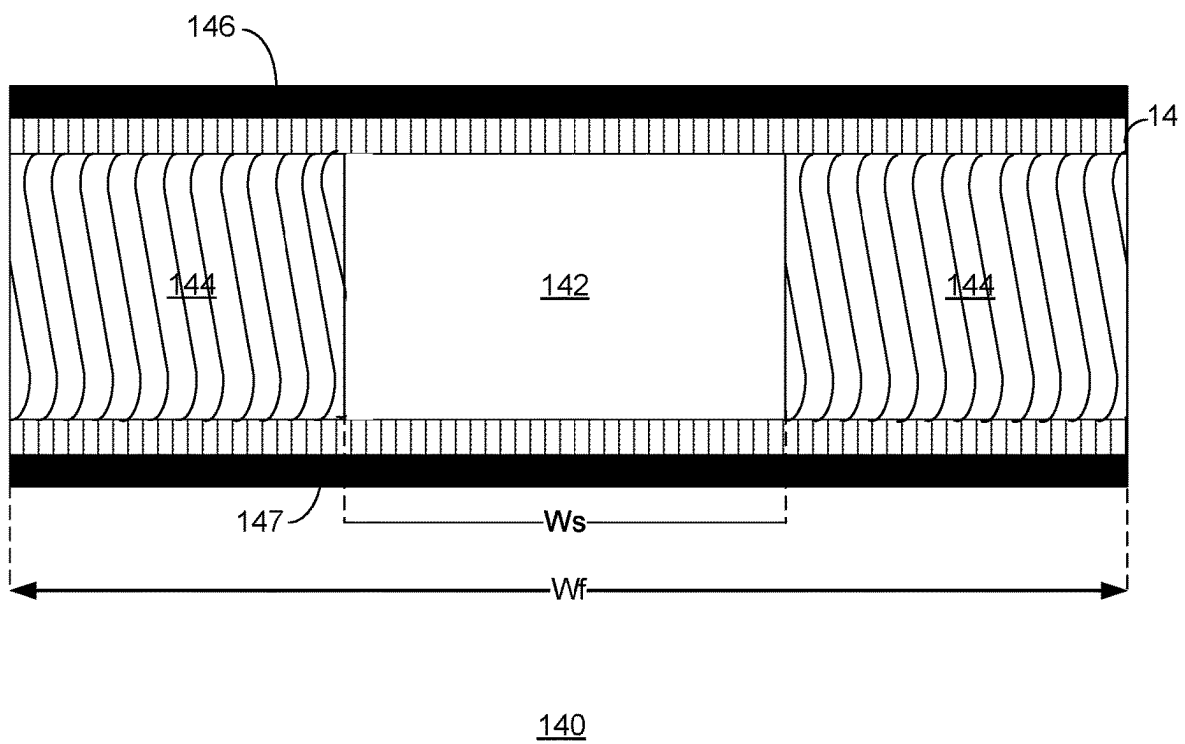


FIG. 1A

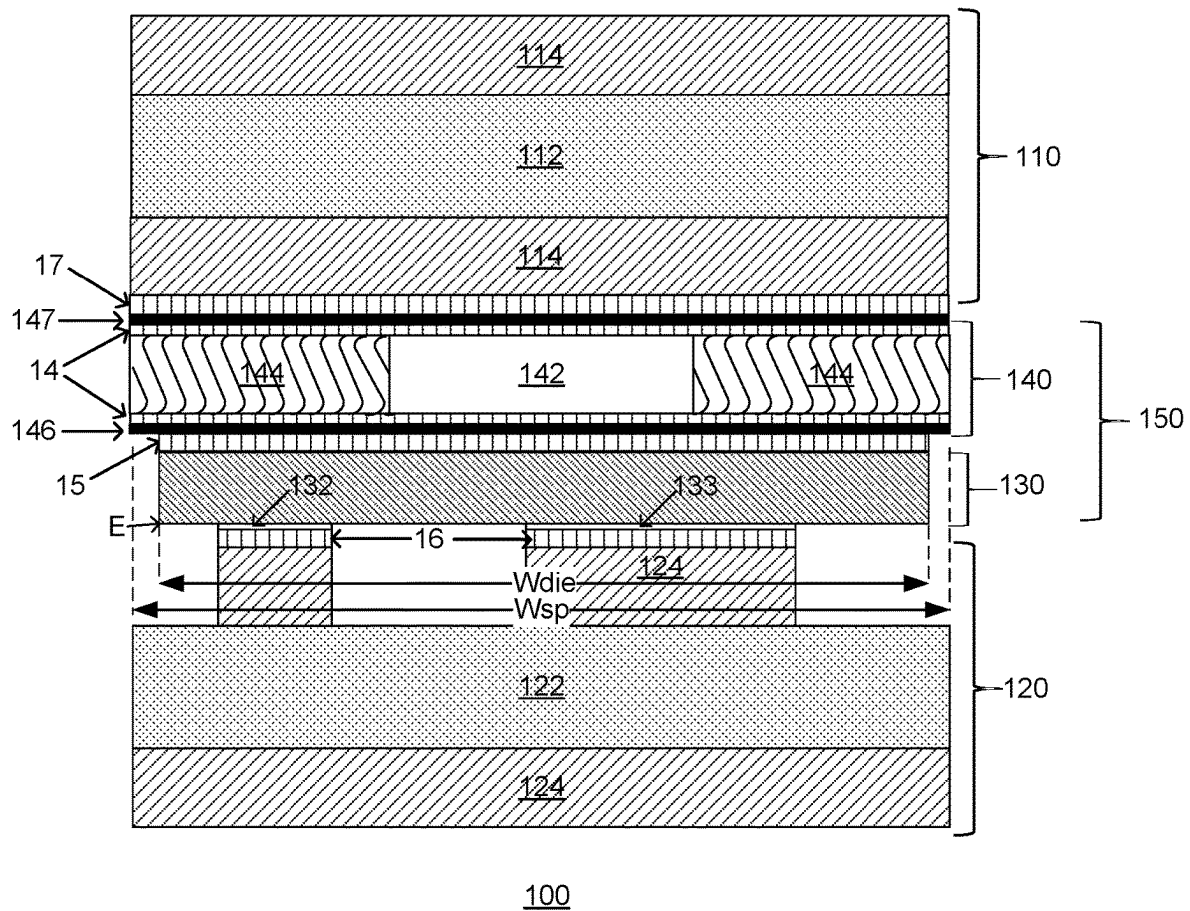


FIG. 1B

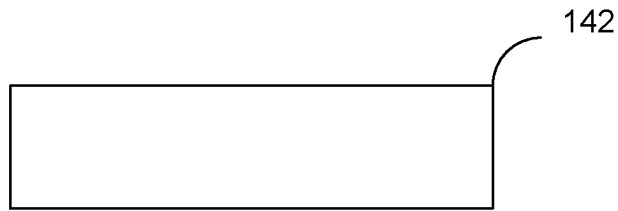


FIG. 2A

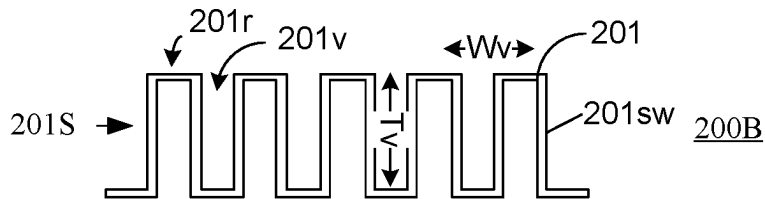


FIG. 2B

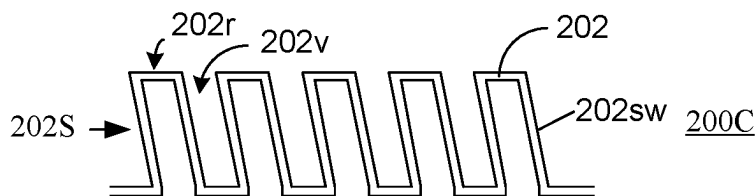


FIG. 2C

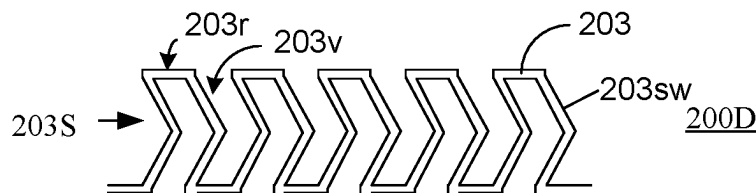


FIG. 2D

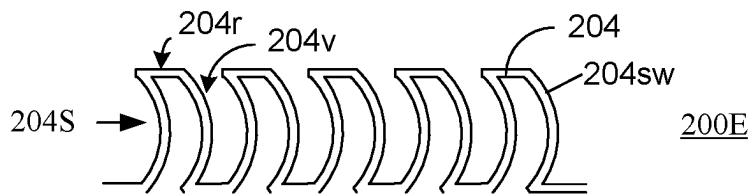


FIG. 2E

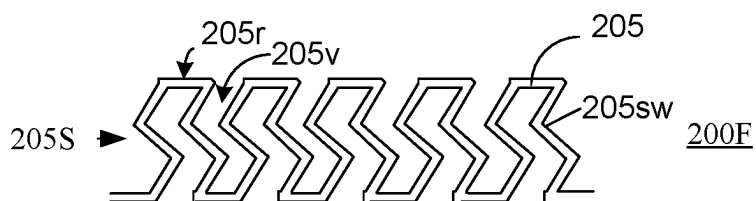


FIG. 2F

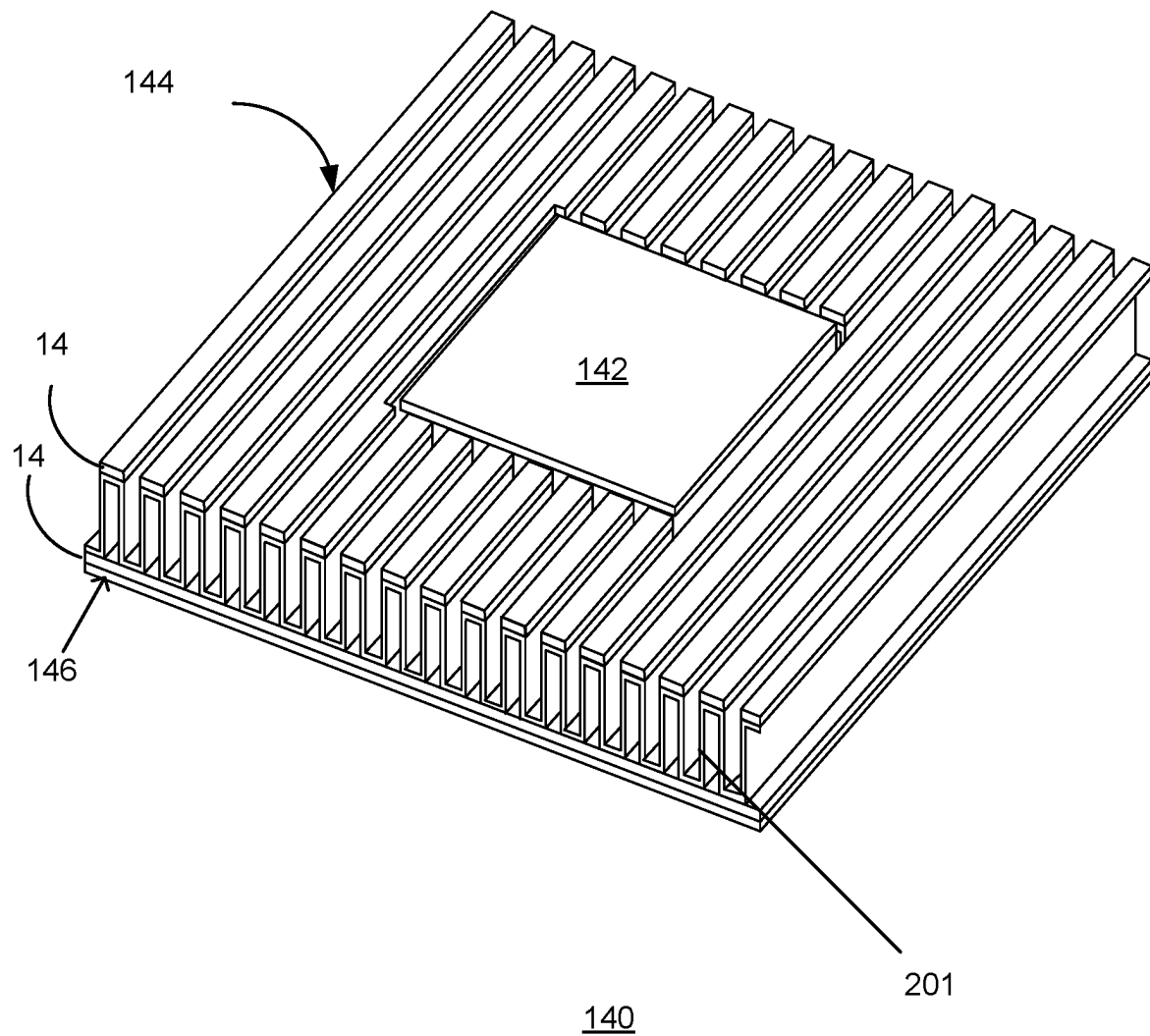


FIG. 3

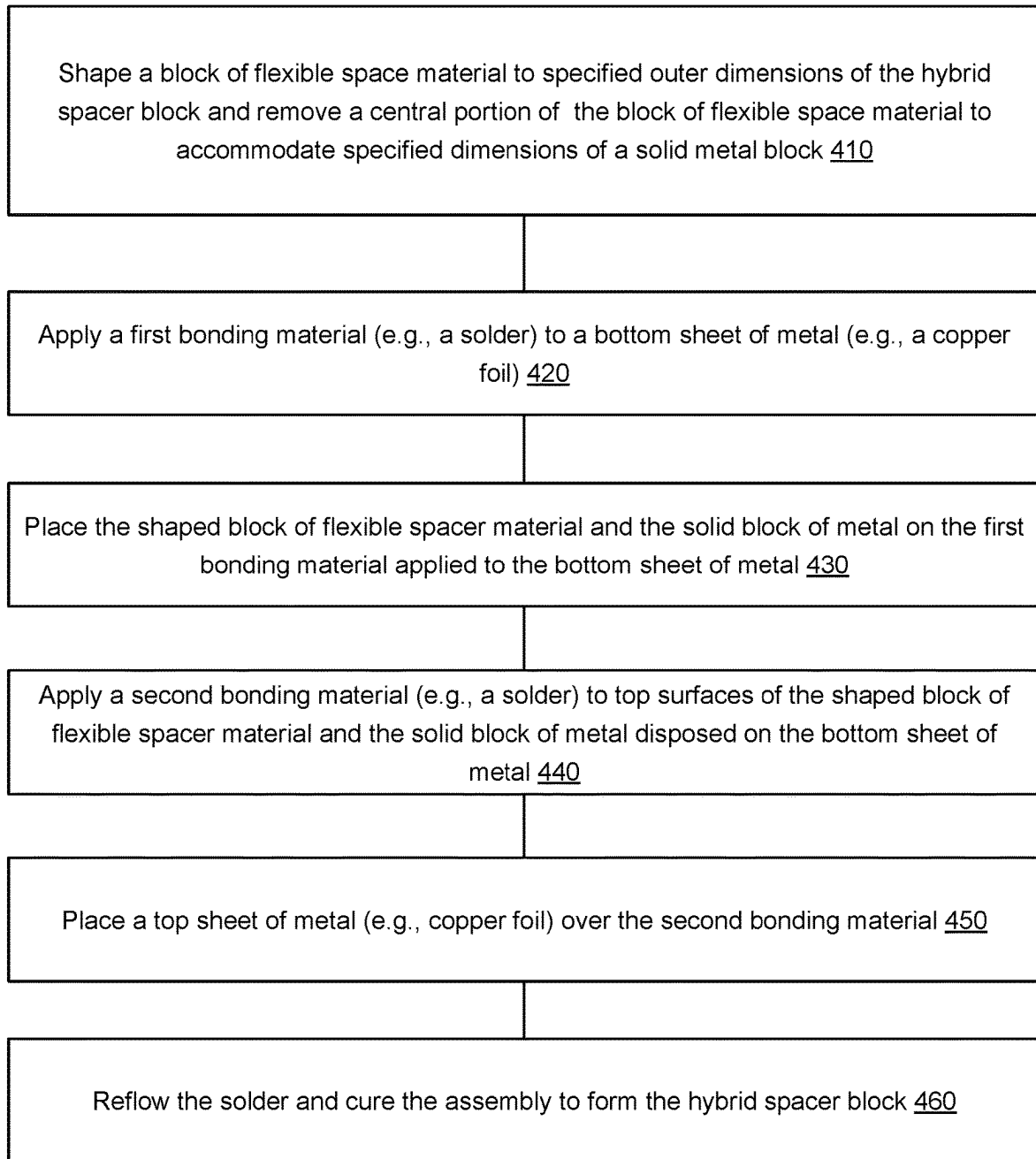
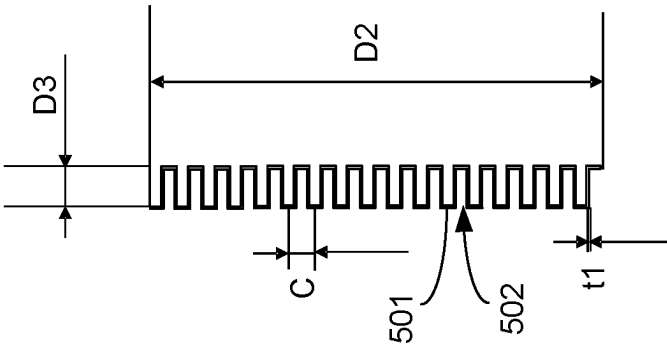
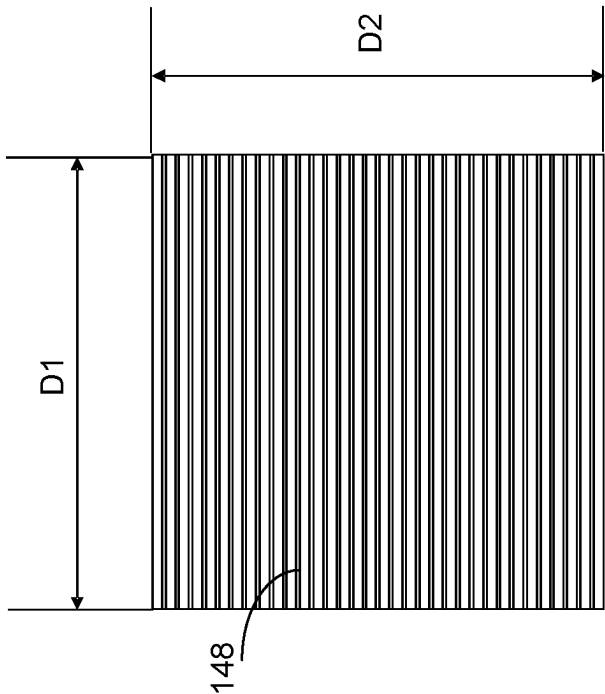
400

FIG. 4



500
FIG. 5B



500
FIG. 5A

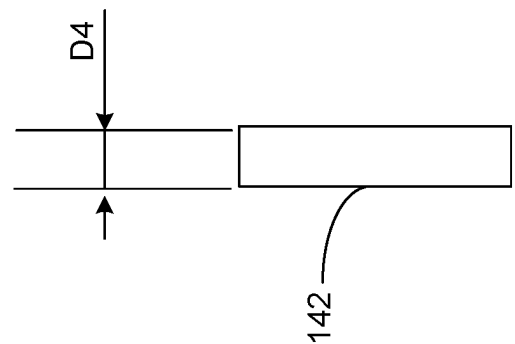


FIG. 5D

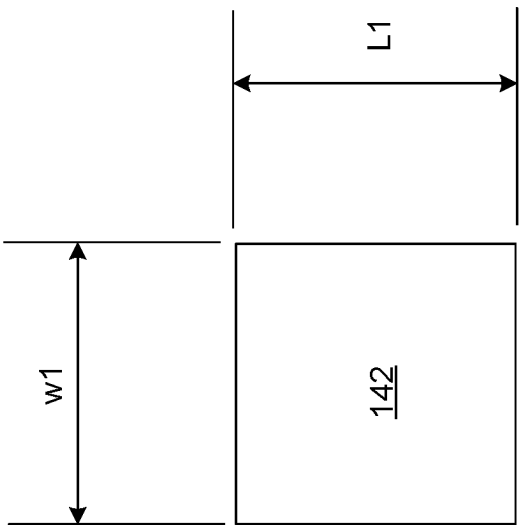


FIG. 5C

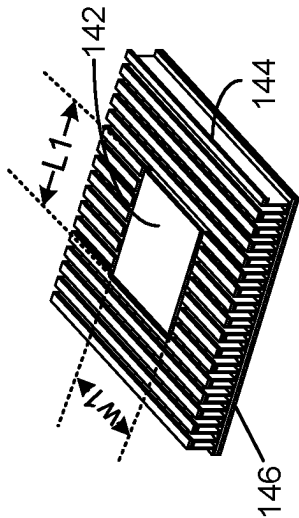


FIG. 5G

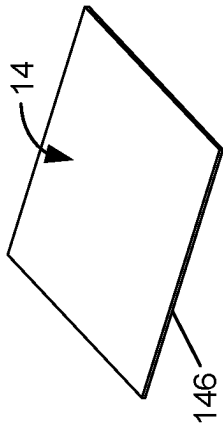


FIG. 5F

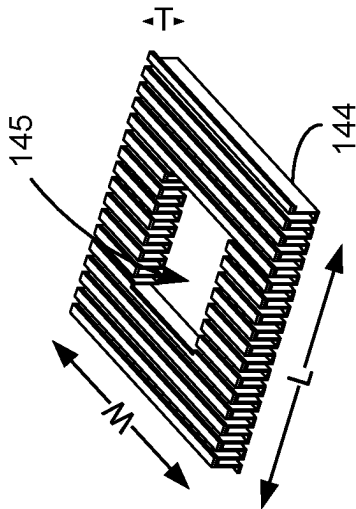


FIG. 5E

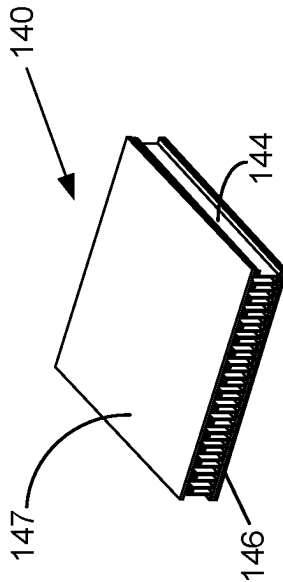


FIG. 5I

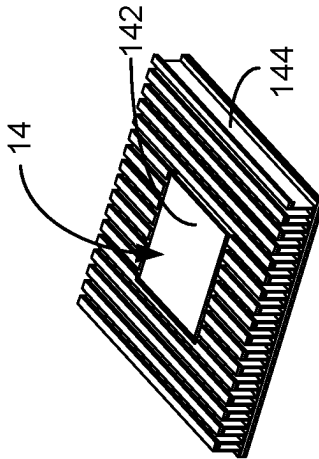
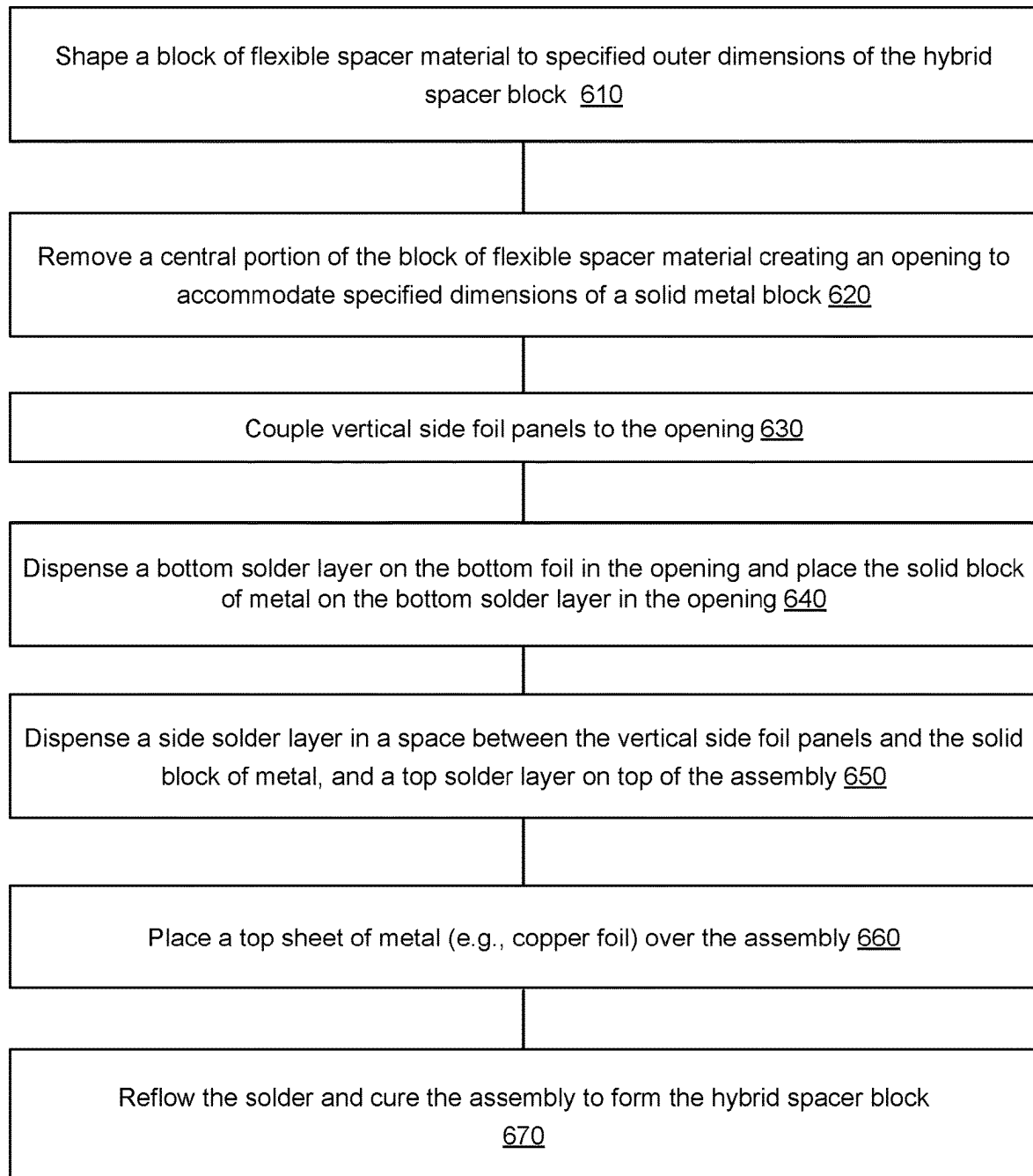


FIG. 5H

600**FIG. 6**

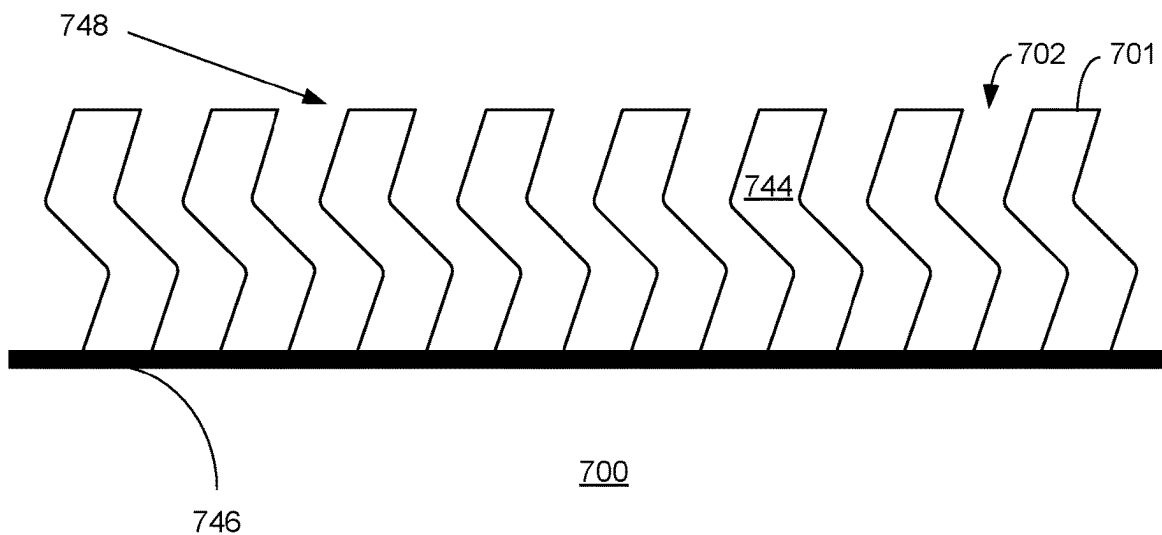


FIG. 7A

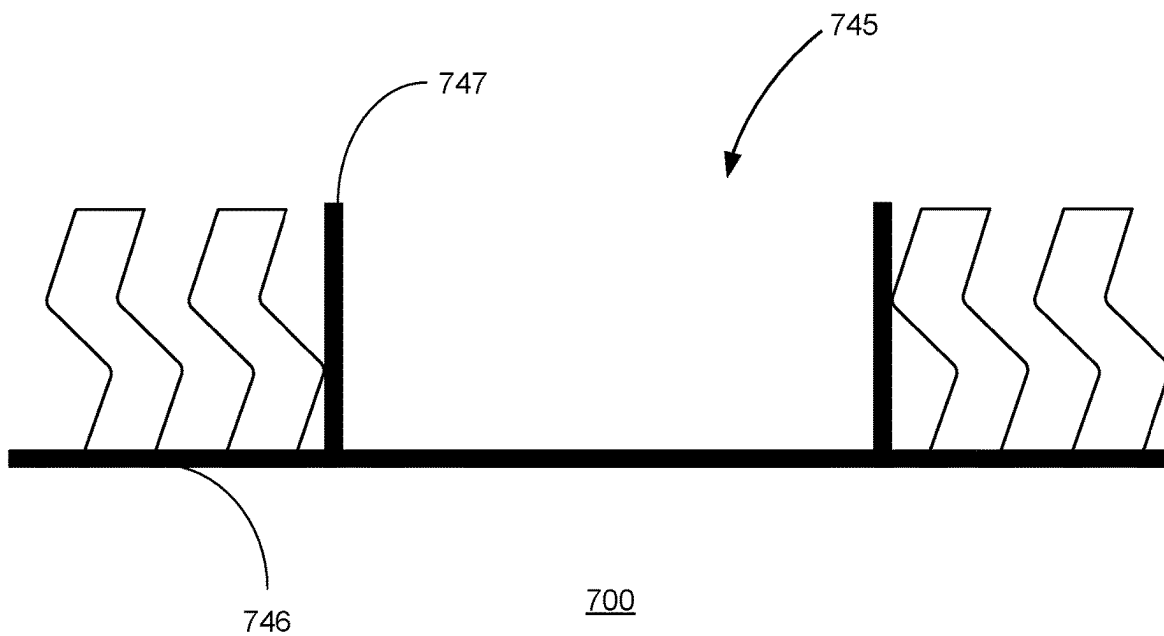


FIG. 7B

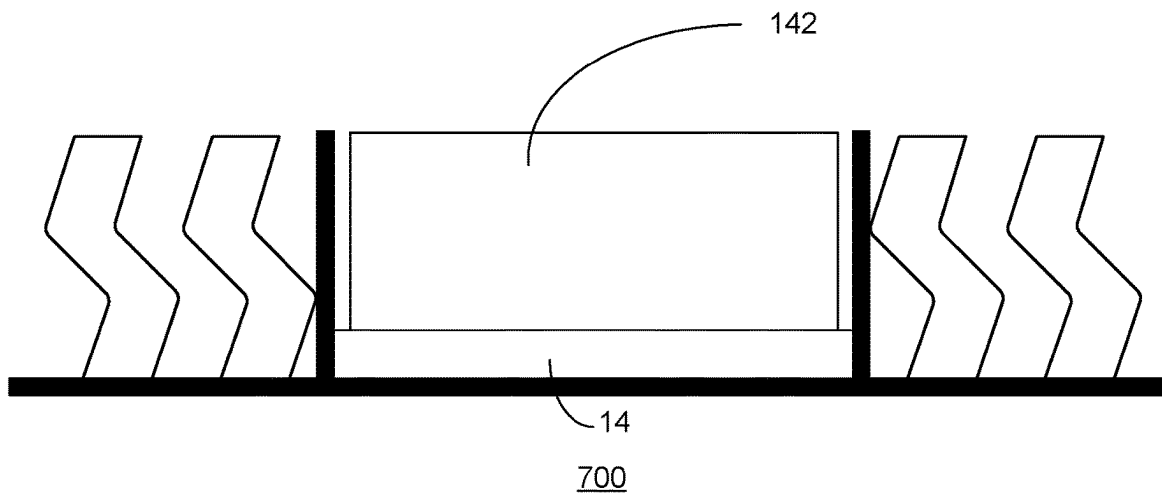


FIG. 7C

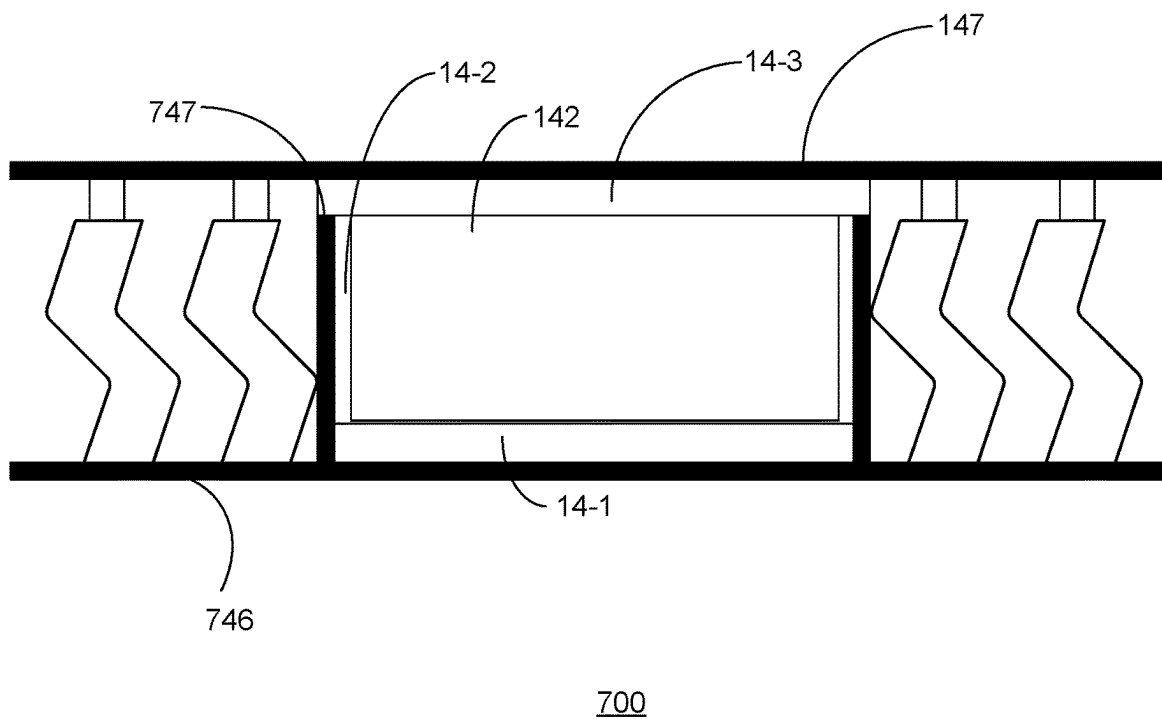


FIG. 7D

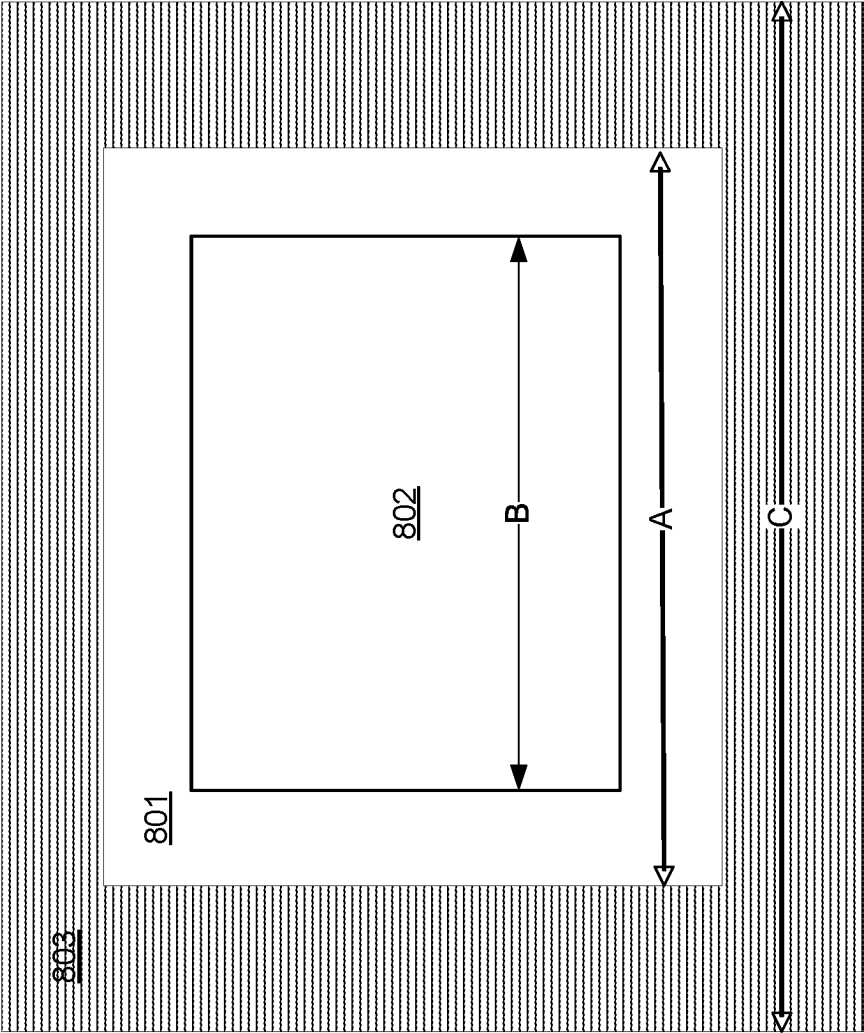
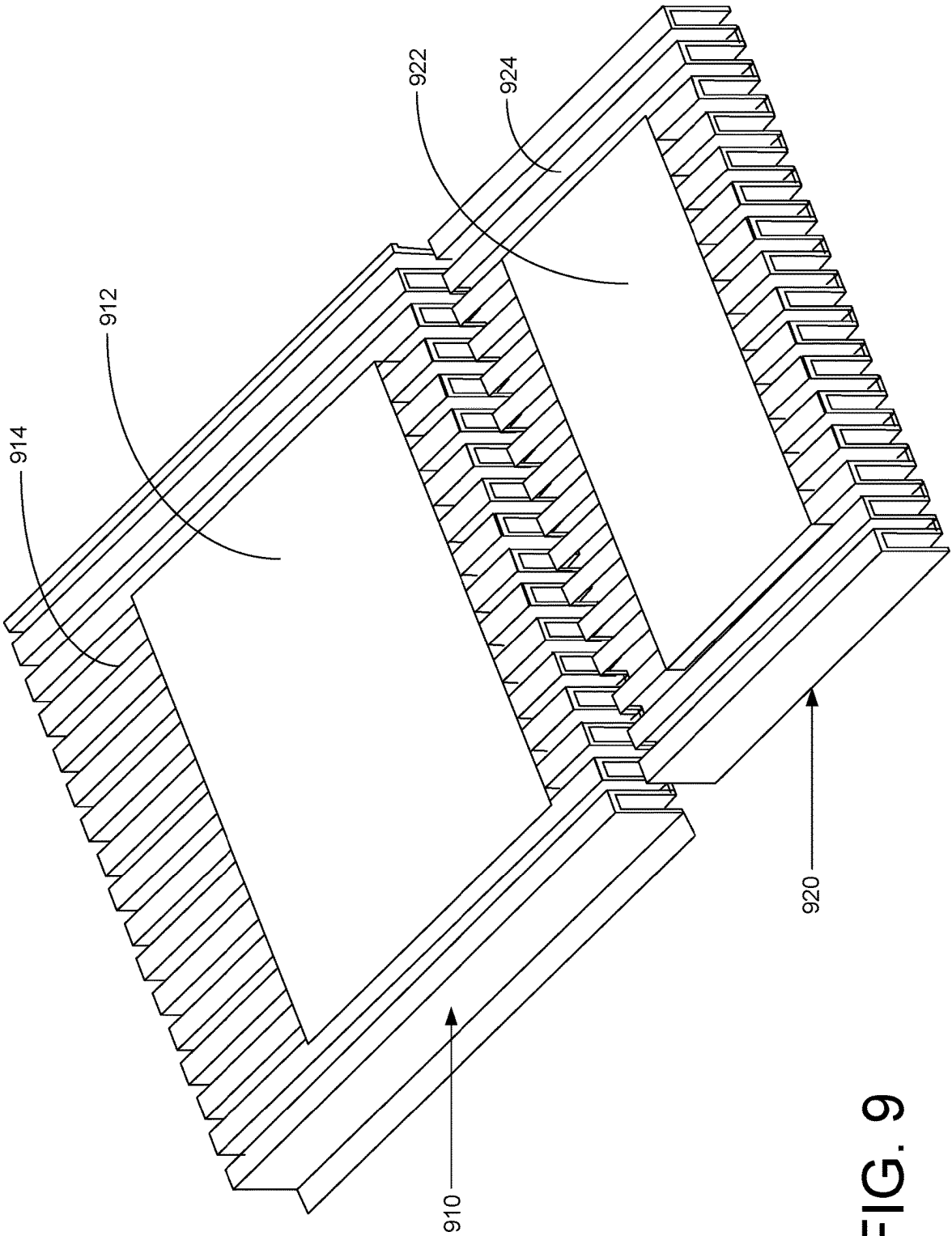


FIG. 8



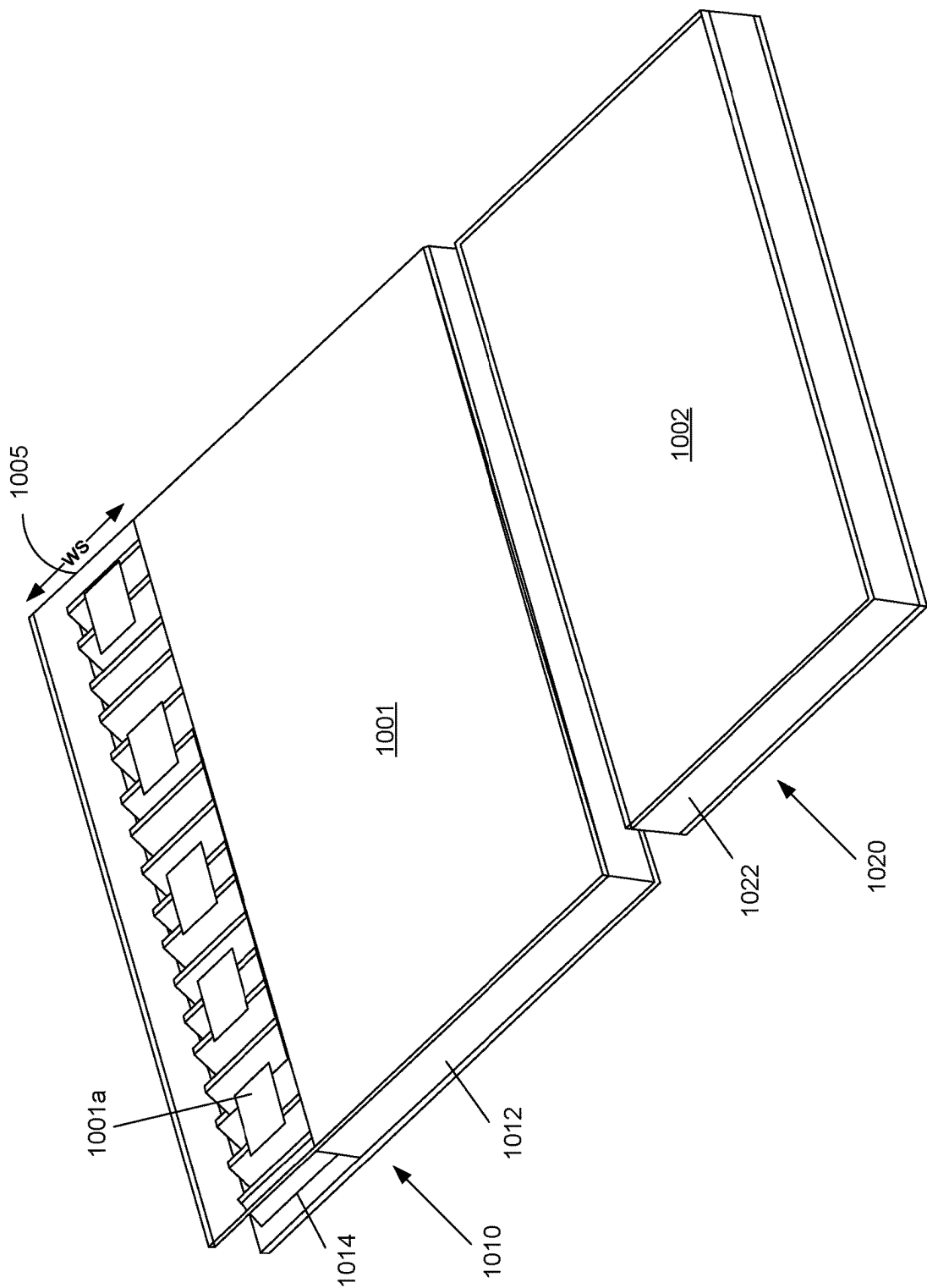


FIG. 10

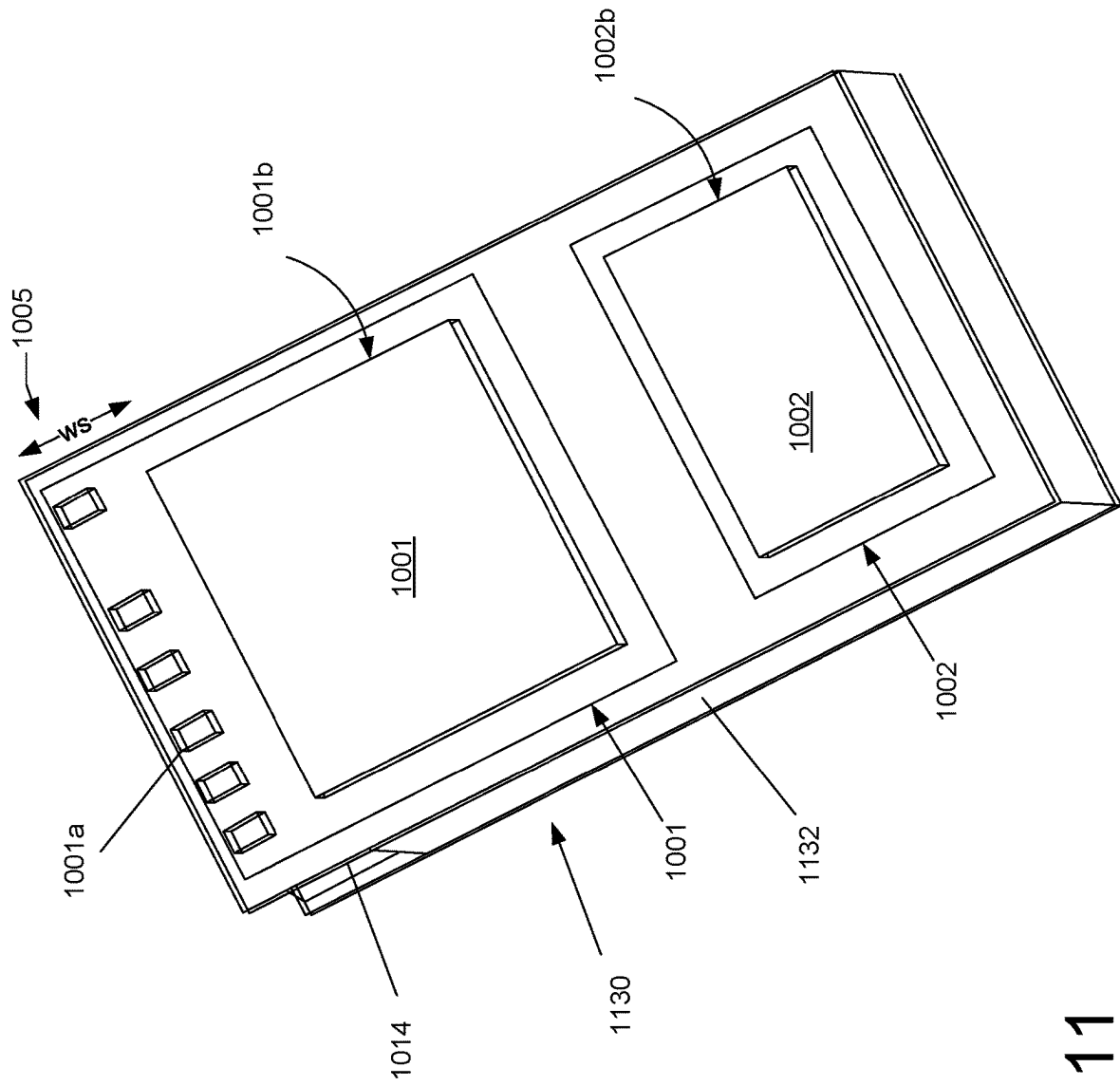


FIG. 11

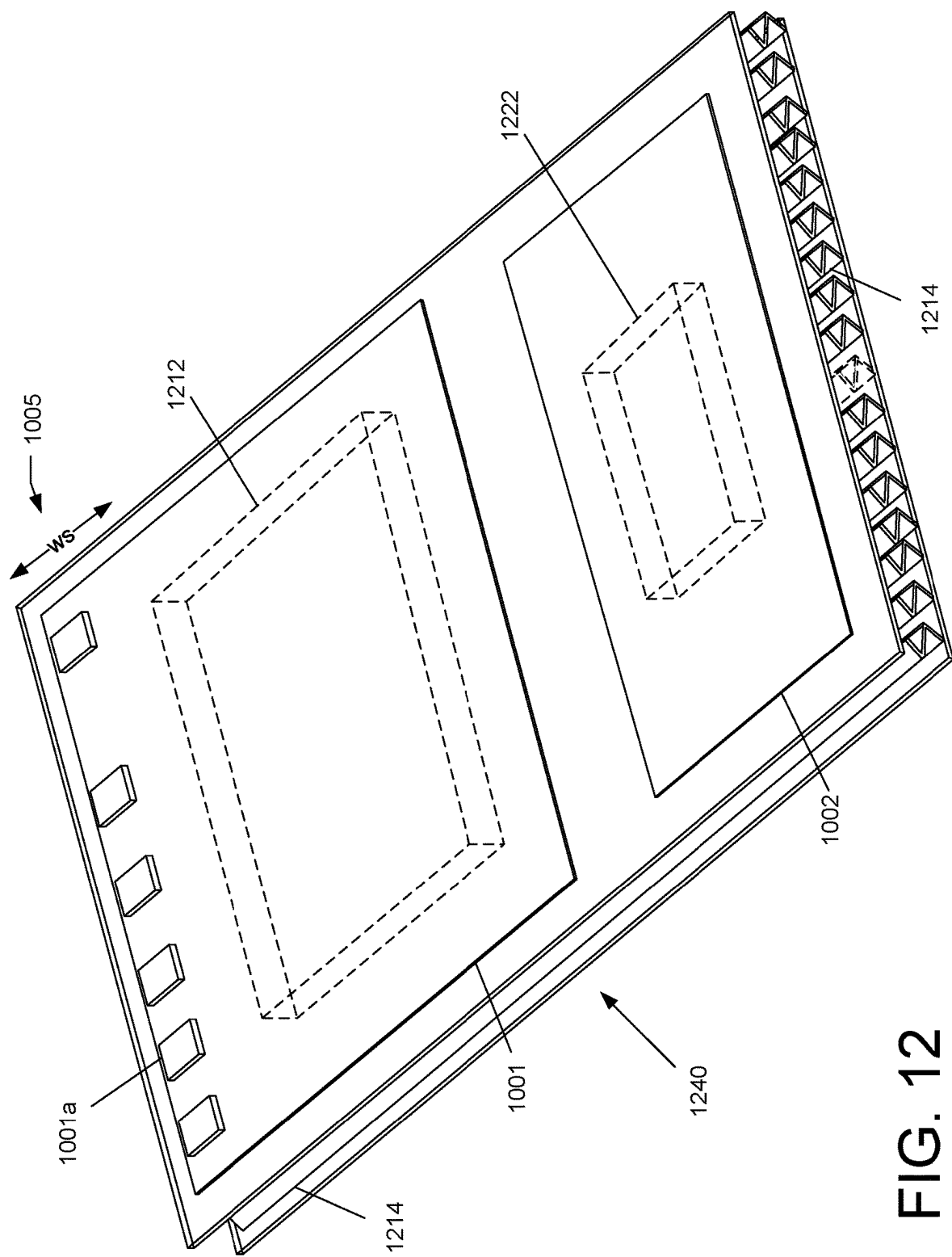
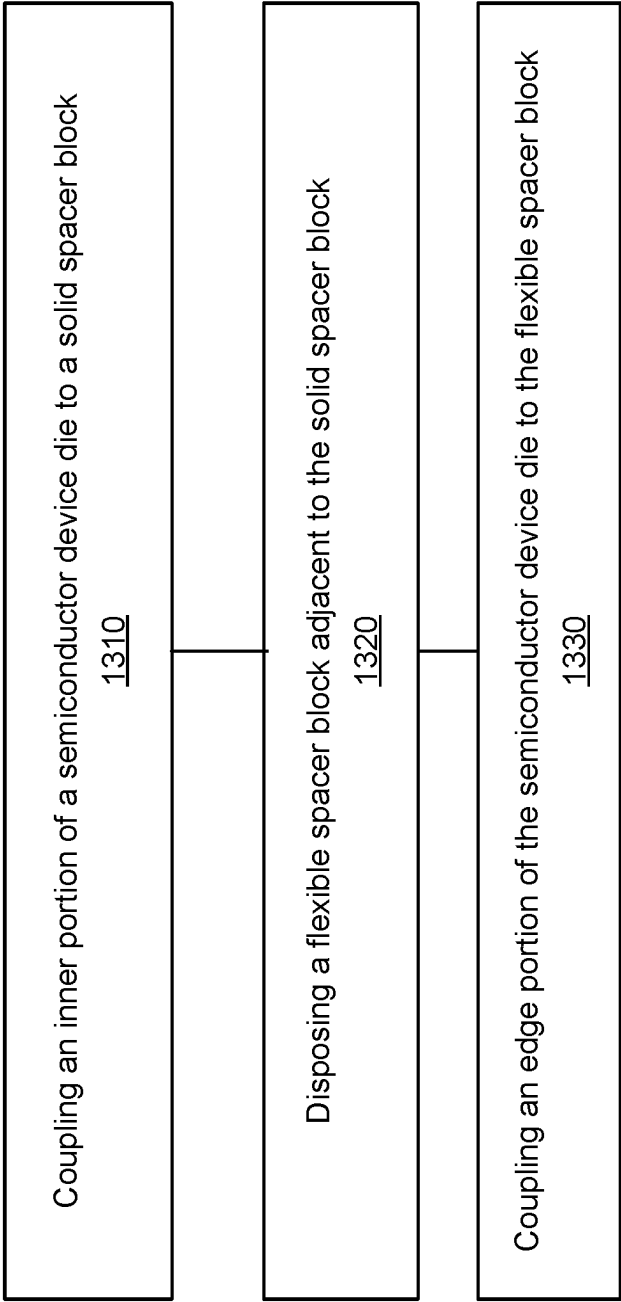


FIG. 12



1300

FIG. 13

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STRUCTURE AND METHOD RELATED TO A POWER MODULE USING A HYBRID SPACER

RELATED APPLICATIONS

This application is a continuation application of U.S. patent application Ser. No. 17/655,398, filed Mar. 18, 2022 as a continuation of U.S. patent application Ser. No. 16/784,999, filed Feb. 7, 2020, both of which are incorporated by reference herein in their entireties.

TECHNICAL FIELD

This description relates to packaging of power devices.

BACKGROUND

Modern high-power devices can be fabricated using advanced silicon technology to meet high power requirements. These high-power devices (e.g., silicon power devices such as an insulated-gate bipolar transistor (IGBT), a fast recovery diode (FRD), etc.) may be packaged in single-side cooling (SSC) or dual-side cooling (DSC) power modules. High-power devices that can deliver or switch high levels of power can be used in, for example, vehicles powered by electricity (e.g., Electric vehicles (EVs), hybrid electric vehicles (HEVs) and plug-in-electric vehicles (PHEV)). The larger size and thicknesses of the high-power device die can create problems such as die warpage and die damage during packaging of the high-power devices for use in circuit packages or power modules (e.g., SSC or DSC power modules), or during stress tests of the fabricated high-power devices.

SUMMARY

In some aspects, the techniques described herein relate to a method, including: shaping a block of flexible spacer material; shaping a portion of the block of flexible spacer material to receive a solid metal block; and coupling the solid metal block to the portion of the block of flexible spacer material.

In some aspects, the techniques described herein relate to a method, wherein the block of flexible spacer material has a corrugated structure made of alternate rectangular ridges and grooves with vertical sidewalls.

In some aspects, the techniques described herein relate to a method, wherein the block of flexible spacer material has a corrugated structure including flexible structures that have non-vertical sidewalls.

In some aspects, the techniques described herein relate to a method, wherein shaping the portion of the block of flexible spacer material to receive the solid metal block includes cutting an opening in the block of flexible spacer material.

In some aspects, the techniques described herein relate to a method, wherein coupling the solid metal block to the portion of the block of flexible spacer material includes disposing the solid metal block in the opening in the block of flexible spacer material.

In some aspects, the techniques described herein relate to a method, further including: applying a first bonding material to a first sheet of metal and placing the block of flexible spacer material including the solid metal block on the first sheet of metal.

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In some aspects, the techniques described herein relate to a method wherein the first sheet of metal is a copper foil.

In some aspects, the techniques described herein relate to a method, further including: applying a second bonding material to a top surface of the block of flexible spacer material including the solid metal block placed on the first sheet of metal and placing a second sheet of metal over the second bonding material.

In some aspects, the techniques described herein relate to a method, wherein the second sheet of metal is a copper foil.

In some aspects, the techniques described herein relate to a method, wherein the first bonding material and the second bonding material are solders, and the method further includes reflowing the solders and heat curing the first bonding material and the second bonding material.

In some aspects, the techniques described herein relate to a method, wherein coupling the solid metal block to the portion of the block of flexible spacer material includes adding side foil panels to the opening.

In some aspects, the techniques described herein relate to a method further including: coupling a semiconductor die to the solid metal block in a semiconductor device module.

In some aspects, the techniques described herein relate to a method, wherein coupling the semiconductor die includes coupling an inner portion of the semiconductor die to the solid metal block, and coupling an outer portion of the semiconductor die to the block of flexible spacer material.

In some aspects, the techniques described herein relate to a method, including: shaping a block made of flexible spacer material, the flexible spacer material including flexible structures that have non-vertical walls; forming an opening in the block of flexible spacer material; disposing vertical side foil panels in the opening; and disposing a solid metal block in the opening in the block of flexible spacer material.

In some aspects, the techniques described herein relate to a method, wherein the flexible spacer material includes a plurality of flexible fins attached to a bottom foil.

In some aspects, the techniques described herein relate to a method further including dispensing a bottom solder layer on the bottom foil in the opening, and disposing the solid metal block on the bottom solder layer in the opening.

In some aspects, the techniques described herein relate to a method further including: dispensing a side solder layer in a space between the vertical side foil panels and the solid metal block.

In some aspects, the techniques described herein relate to a method further including: disposing a top solder layer on the block of flexible spacer material including the solid metal block.

In some aspects, the techniques described herein relate to a method further including reflowing the bottom solder layer, the top solder layer, and the side solder layer.

In some aspects, the techniques described herein relate to a method, including: coupling a flexible spacer block and a solid metal block to form a hybrid spacer block; coupling an inner portion of a semiconductor die to the solid metal block; and coupling an outer portion of the semiconductor die to the flexible spacer block, wherein the flexible spacer block is a strip attached to, and extending from, a side of the solid metal block toward an edge of the semiconductor die.

In some aspects, the techniques described herein relate to a method, wherein the flexible spacer block has a corrugated structure made of at least one of a straight fin, a tilted fin, a bent fin, a curved fin, or an S-shaped fin.

In some aspects, the techniques described herein relate to a method further including: bonding the hybrid spacer block

to a substrate, the hybrid spacer block providing a thermal pathway for heat generated in the semiconductor die to the substrate.

The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a block diagram illustrating an example hybrid spacer block.

FIG. 1B is a block diagram illustrating an example power module utilizing a hybrid spacer block for a die-supporting spacer block combination.

FIG. 2A illustrates a solid block.

FIGS. 2B through 2F illustrate different types of flexible spacer configurations that can be used in a hybrid spacer block.

FIG. 3 illustrates an example hybrid spacer block.

FIG. 4 illustrates an example method for assembling a hybrid spacer block.

FIGS. 5A through 5I illustrate a hybrid spacer block at different stages of assembly by the method of FIG. 4.

FIG. 6 illustrates an example method for assembling a hybrid spacer block.

FIGS. 7A through 7D illustrate a hybrid spacer block at different stages of assembly by the method of FIG. 6.

FIG. 8 shows schematic depicting relative sizes of an individual die, a solid spacer block, and a flexible spacer block.

FIG. 9 is a schematic illustration of hybrid spacer blocks for use in a multi-die power module.

FIG. 10 is a schematic illustration an arrangement of hybrid spacer blocks that may be individually coupled to a first device die and a second device die in a multi-die power module.

FIG. 11 is a schematic illustration of a hybrid spacer block coupled to a first device die and a second device die in a multi-die power module.

FIG. 12 is a schematic of a hybrid spacer block coupled to a first device die and a second device die in a multi-die power module.

FIG. 13 illustrates an example method of using a flexible spacer block in a power module.

DETAILED DESCRIPTION

Modern high-power semiconductor devices can be fabricated using advanced silicon technologies to meet the high power requirements. The power devices (e.g., an insulated-gate bipolar transistor (IGBT), a fast recovery diode (FRD), etc.) may be fabricated using, for example, one or more of silicon (Si), silicon carbide (SiC), and gallium nitride (GaN) materials, or other semiconductor materials. The power devices may be fabricated on thinned semiconductor wafers (e.g., silicon wafers) that are, for example, only about 100 microns thick or less. This results in high-power device die sizes that are larger and thinner than power device die sizes for traditional power devices fabricated on regular substrates (i.e., on un-thinned silicon wafers) using conventional silicon technologies.

In circuit packages (e.g., a dual-side cooling (DSC) power module), the semiconductor device die can be placed between a pair of direct bonded copper (DBC) substrates that help conduct heat away from the semiconductor device die. The individual thinned device die can be mechanically

and structurally reinforced by bonding (e.g. soldering) a supporting spacer block (i.e., a thermally conductive spacer block) to a backside of the die. The spacer also provides good electrical isolation and thermal performance. The other side of the supporting spacer block can be bonded to a DBC substrate. The supporting spacer block provides a thermal pathway for heat flow from the semiconductor device die to the DBC substrate. In a DSC power module, the die-supporting block combination is placed between the pair of DBC substrates and encapsulated in molding material (e.g., epoxy molding compound (EMC)). In instances in which the die is placed in a flip-chip configuration on a DBC substrate, a molded underfill (MUF) molding material can be introduced to underfill a narrow gap under the flip-chip die, and over molding the die-supporting block combination for encapsulation.

In some implementations, the supporting spacer block is a solid piece of thermally conductive material (e.g., a rectangular solid copper block having a cross sectional area A and a thickness T). The thickness of the thin device die is augmented by the thickness of the solid copper block in the die-supporting spacer block combination. The solid copper block can have a smaller cross sectional area than the area of device die or can have a larger cross sectional area that may be comparable to the area of the device die. However, cracks in the die can form around the solid copper block when the block area is smaller than die area. Use of solid copper blocks that have larger areas (e.g., greater or comparable to the area of the device die) has been proposed as a remedy, but large area solid copper blocks also result in die cracking. Solder voids may form between a larger area solid copper block and die in the die-supporting spacer block combination. These solder voids can induce die corner cracks at die corners.

A cause of the die corner cracks may be a coefficient of thermal expansion (CTE) mismatch between the spacer block (e.g., a copper block) and the MUF (or EMC) used in the die-supporting spacer block combination. As the copper spacer block shrinks faster than the MUF (or EMC) on cooling, the MUF (or EMC) will oppose the die motion induced by shrinkage of the solid copper spacer block at a die corner with the solder voids and lead to stress that creates the die corner cracks.

In accordance with the principles of the present disclosure, a hybrid spacer block for a die-supporting spacer block combination has a hybrid structure that includes a solid block spacer portion and a mechanically elastic or flexible block spacer portion. The solid spacer block portion may have a cross sectional area that is smaller than the die area. The solid spacer block portion may be an inner portion (central portion) of the hybrid spacer block. The flexible block spacer portion may be disposed around (e.g., surrounding, or adjacent) the solid spacer block portion. The flexible spacer block portion and the solid spacer block portion in combination may have a cross sectional area that is comparable to or greater than the die area. In the die-supporting spacer block combination, the flexible spacer block portion may extend to the die corners and may flexibly accommodate any mechanical displacements due to CTE mismatch between the components (e.g., metal, EMC and die semiconductor). This flexible accommodation of mechanical displacements due to CTE mismatch may prevent die cracking at die corners, and reduce instances of solder peeling failure under the device pads (e.g., signal pads) in a DSC power module that utilizes the die-supporting spacer block combination.

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FIG. 1A shows a cross sectional view of an example hybrid spacer block **140**, in accordance with the principles of the present disclosure. The hybrid spacer block **140** may include a solid metal block **142** (e.g., a solid copper block) that is surrounded (e.g., entirely surrounds, at least partially surrounds) by a flexible spacer block **144** (e.g., a mechanically flexible copper block). Solid metal block **142** may have a width W_s . Hybrid spacer block **140** including flexible spacer block **144** may have width W_f . Solid metal block **142** and flexible spacer block **144** may be disposed between metal sheets **146** and **147** (e.g., copper sheets or foils) and held in place between metal sheets **146** and **147** by solder (e.g., solder **14**).

FIG. 1B shows a cross sectional view of an example DSC power module **100** utilizing a hybrid spacer block **140** in a die-supporting spacer block combination, in accordance with the principles of the present disclosure. The example DSC power module **100** can utilize the hybrid spacer block **140** discussed in connection with at least FIG. 1A.

DSC power module **100** may enclose a device die **130** and a hybrid spacer block **140** between two DBC substrates (e.g., DBC substrate **110** and DBC substrate **120**). On one side the hybrid spacer block (e.g., hybrid spacer block **140**) is coupled to the device die **130**, and on another side the hybrid spacer block is bonded to one of the two DBC substrates (e.g., DBC substrate **110**). The hybrid spacer block provides a thermal pathway for heat generated in the device die to be dissipated over DBC substrate **110**.

Hybrid spacer block **140** may include a solid metal block **142** (e.g., a solid copper block) that is surrounded by a flexible spacer block **144** (e.g., a mechanically flexible copper block). Solid metal block **142** and flexible spacer block **144** may be disposed between metal sheets **146** and **147** (e.g., copper sheets or foils) and held in place between metal sheets **146** and **147** by solder (e.g., solder **14**). Device die **130** (e.g., an IGBT device) may be bonded (e.g., soldered) to hybrid spacer block **140** via a solder layer **15** forming a die-spacer block combination **150** of DSC power module **100**. An inner portion of device die **130** may be bonded to solid metal block **142**. An outer portion of the device die (e.g., extending from the inner portion to an edge E of the device die) may be bonded to the flexible spacer block **144**. In example implementations, as shown in FIG. 1B, hybrid spacer block **140** may have a width W_{sp} that is about a same or greater than a width W_{die} of device die **130**. In this geometrical arrangement, any displacement of device die edge E (e.g., device die bending induced by thermal mismatch of components) can be accommodated (e.g., mechanically) by the elasticity of flexible spacer block **144** and die cracking can be prevented.

In example implementations, as shown in FIG. 1B, DBC substrate **110** and DBC substrate **120** may be ceramic tiles (e.g., alumina tiles **112**, **122**) that are plated with copper layers (e.g., layers **114**, **124**). Device die **130** (e.g., an IGBT device) may have one or more contact pads (e.g., signal pads **132** and source pads **133**).

In example implementations, as shown in FIG. 1B, DSC power module **100** may be assembled, for example, with die-spacer block combination **150** in an orientation that has device **130** in a flip chip configuration. Device **130** in the flip chip configuration may have signal pads **132** and source pads **133** bonded to DBC substrate **120** via a solder layer **16**. Further, hybrid spacer block **140** (of die-spacer block combination **150**) may be bonded to DBC substrate **110** via a solder layer **17**.

The mechanically elastic or flexible spacer material used for flexible spacer block **144** in hybrid spacer block **140** can

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be of different types. FIGS. 2B through 2F show cross sectional views of different types of flexible spacer configurations that may be used in hybrid spacer block **140**. FIG. 2A shows, for comparison, a cross sectional view of solid metal block **142**.

FIG. 2B illustrates example flexible spacer material **200B** that includes a corrugated structure **201S** made of straight fins (e.g., straight fin **201**). As pictorially shown in FIG. 2B, the straight fins may be arranged so that corrugated structure **201S** is made of alternate rectangular ridges **201r** and grooves or valleys **201v** with vertical sidewalls **201sw** formed by the straight fins **201**. Ridges **201r** and valleys **201s** may alternate with a spatial periodicity W_v . Valleys **201v** with vertical sidewalls **201sw** may have a height T_v .

FIG. 2C illustrates example flexible spacer material **200C** that includes a corrugated structure **202S** made of tilted fins (e.g., tilted fin **202**). As pictorially shown in FIG. 2C, the tilted fins may be arranged so that corrugated structure **202S** is made of alternate ridges **202r** and grooves or valleys **202v** with slanted or tilted sidewalls **202sw** formed by the tilted fins **202**.

FIG. 2D illustrates example flexible spacer material **200D** that includes a corrugated structure **203S** made of bent fins (e.g., bent fin **203**). As pictorially shown in FIG. 2C, the bent fins may be arranged so that corrugated structure **203S** is made of alternate ridges **203r** and grooves or valleys **202v** formed by bent fins **203** with bent sidewalls **203sw**.

FIG. 2E illustrates example flexible spacer material **200E** that includes a corrugated structure **204S** made of curved fins (e.g., curved fin **204**). As pictorially shown in FIG. 2E, the curved fins may be arranged so that corrugated structure **204S** is made of alternate rectangular ridges **204r** and grooves or valleys **204v** formed by curved fins **204** with curved sidewalls **204sw**.

FIG. 2F illustrates example flexible spacer material **200F** that includes a corrugated structure **205S** made of S-shaped fins (e.g., S-shaped fin **205**). As pictorially shown in FIG. 2F, the S-shaped fins may be arranged so that corrugated structure **205S** is made of alternate rectangular ridges **205r** and grooves or valleys **205v** formed by S-shaped fins **204** with S-shaped sidewalls **205sw**.

The different types of flexible spacer material with the different types of fin structure (e.g., straight fin **101**, tilted fin **202**, bent fin **203**, curved fin **204**, and S-shaped fin **205**, etc.) may impart different mechanical and thermal properties to flexible spacer block **144**.

FIGS. 1A and 1i, as discussed above, show, for example, an implementation in which flexible spacer block **144** is made of flexible spacer material of the S-shaped fin type (i.e., S-shaped fin **205**).

FIG. 3 shows, for example, an implementation in which hybrid spacer block **140** uses material of the straight fin type (i.e., straight fin **201**) for flexible spacer block **144**. As shown in FIG. 3, in hybrid spacer block **140**, flexible spacer block **144** may surround and be coupled to a solid spacer block (e.g., solid metal block **142**).

FIG. 4 shows an example method **400** for assembling a hybrid spacer block (e.g., hybrid spacer block **140**) for use in die-spacer block combination in a power module.

Method **400** includes shaping a block of flexible spacer material to specified outer dimensions of the hybrid spacer block and preparing an inner portion of the block of flexible spacer material to accommodate specified dimensions of a solid metal block (**410**). The specified outer dimensions of the hybrid spacer block may match the dimensions of the semiconductor die in the die-spacer block combination.

Preparing the inner portion of the block of flexible spacer material to accommodate specified dimensions of a solid metal block **410** may include removing a portion of the block of flexible spacer material to create an opening (e.g., by cutting out a hole) in the block of flexible spacer material. Preparing the inner portion may include adding side foil panels (e.g., copper foil) to the opening or hole.

Method **400** may further include applying a first bonding material (e.g., a solder) to a bottom sheet of metal (e.g., a copper foil) (**420**), placing the shaped block of flexible spacer material and the solid block of metal on the first bonding material applied to the bottom sheet of metal (**430**). Method **400** further includes applying a second bonding material (e.g., a solder) to top surfaces of the shaped block of flexible space material and the solid block of metal disposed on the bottom sheet of metal (**440**), placing a top sheet of metal (e.g., copper foil) over the second bonding material (**450**), and reflowing the solder and curing the assembly to form the hybrid spacer block (**460**).

FIGS. **5A** through **5I** show the hybrid spacer block (e.g., hybrid spacer block **140**) at different stages of assembly by method **400** starting with a block of flexible spacer material **500**.

FIGS. **5A** and **5B** show a starting block of flexible spacer material **500** that can be utilized to make hybrid spacer block **140** according to method **400**. In an example implementation, flexible spacer material **500** may have a straight fin structure, i.e., a corrugated structure **148** made of alternate rectangular ridges **501** and grooves **502** with vertical sidewalls.

FIG. **5A** shows a top view of flexible spacer material **500** having, for example, rectangular slab dimensions $D1 \times D2$ and a thickness $D3$. In example implementations, the dimensions $D1$ and $D2$ each may be greater than about 10 millimeters, and thickness $D3$ may be in the range of 0.5 to 3.0 millimeters.

FIG. **5B** shows a cross sectional view of starting block of flexible spacer material **500** depicting the ridges and grooves of corrugated structure **148** having a thickness $D3$. Corrugated structure **148** may be formed, for example, by rippling a thin metallic sheet (e.g., a copper sheet) having a thickness $t1$ to form ridges **501** and grooves **502**. In example implementations, the thickness $t1$ may be in the range of 0.05 to 3.0 millimeters. Each cell of the corrugated structure (i.e., ridge-to ridge distance) may have a spacing C . In example implementations, spacing C may be in the range of 0.5 to 2 millimeters.

FIGS. **5C** and **5D** show a block of solid material (e.g., solid metal block **142**) that can be used in hybrid spacer block **140**, according to method **400**. FIG. **5C** shows a top view of solid metal block **142** having, for example, rectangular slab dimensions $W1 \times L1$. FIG. **5D** shows a side view of solid metal block **142** having, for example, a thickness $D4$. In example implementations, the dimensions $W1$ and $L1$ each may be smaller than about 50 millimeters, and thickness $D4$ may be in the range of 0.5 to 3.0 millimeters.

FIG. **5E** shows flexible spacer block **144** that has been shaped (from flexible spacer material **500**) to specified outer dimensions of the hybrid spacer block (e.g., at **410**, method **400**). Flexible spacer block **144** may, for example, have a width W , a length L , and a thickness T that correspond to specified outer dimensions of the hybrid spacer block **140**. Further, an open inner portion **145** of flexible spacer block is cut out to accommodate specified dimensions of solid metal block **142** of hybrid spacer block **140**. Inner portion **145** may, for example, have rectangular dimensions $L1 \times W1$ matching the dimension of solid metal block **142** (FIG. **5C**).

FIG. **5F** shows a first bonding material (e.g., solder **14**) being applied to a metal sheet **146** that may form a bottom support for flexible spacer block **144** and solid metal block **142** in hybrid spacer block **140** (e.g., at **420**, method **400**).

FIG. **5G** shows flexible spacer block **144** and solid metal block **142** assembled on bottom metal sheet **146** with solid metal block **142** placed in a hole (e.g., open inner portion **145**) in flexible spacer block **144** (e.g., at **430**, method **400**).

FIG. **5H** shows a second bonding material (e.g., a solder **14**) applied to top surfaces of the shaped block of flexible spacer material and the solid block of metal disposed on the bottom metal sheet **146** (e.g., at **440**, method **400**).

FIG. **5I** shows a top sheet of metal (e.g., metal sheet **147**) placed over the second bonding material of the assembly of FIG. **5F** (e.g., at **450**, method **400**). The assembly may be heated for solder reflow and curing to bond the top and bottom metal sheets **146**, **147** to the assembly components and, thus, form hybrid spacer block **140**.

In an example implementations discussed above with reference to FIGS. **5A-5I**, flexible spacer material **500** has a straight fin structure (i.e., a corrugated structure **148** made of alternating rectangular ridges **501** and grooves **502** with vertical sidewalls). In other implementations, flexible spacer material **500** may have other types of flexible structures with non-vertical sidewalls (e.g., tilted fin **202**, bent fin **203**, curved fin **204**, and S-shaped fin **205**, etc., shown in, for example, FIGS. **2B** through **2F**).

FIG. **6** shows an example method **600** for assembling a hybrid spacer block (e.g., hybrid spacer block **140**) using starting flexible materials that have flexible structures that have non-vertical walls (e.g., a structure with S-shaped fin **205**). In such spacer materials, the flexible fins (e.g., S fins) may be pre-attached to a bottom foil.

Method **600** includes shaping a block of flexible spacer material to specified outer dimensions of the hybrid spacer block (**610**) and removing an inner portion of the block of flexible spacer material to create an opening that can accommodate specified dimensions of a solid metal block (**620**), and coupling (e.g., adding) vertical side foil panels to the opening (**630**).

Method **600** further includes dispensing a bottom solder layer on the bottom foil in the opening and placing the solid block of metal on the bottom solder layer in the opening (**640**). Method **600** further includes dispensing a side solder layer in a space between the vertical side foil panels and the solid block of metal, and a top solder layer on top of the assembly (**650**), and placing a top sheet of metal (e.g., copper foil) over the assembly (**660**), and reflowing the solder and curing the assembly to form the hybrid spacer block (**670**).

FIGS. **7A** through **7D** show the hybrid spacer block (e.g., hybrid spacer block **140**) at different stages of assembly by method **600** starting with a block of flexible spacer material with S-shaped fins.

FIG. **7A** shows a cross sectional view a block of flexible spacer material **700** shaped to specified outer dimensions of the hybrid spacer block. Flexible spacer material **700** may have a S-shaped fin structure (i.e., a corrugated structure **748** made of alternate ridges **701** and grooves **702** formed by S-shaped fins **744** with non-vertical vertical sides). The S-shaped fins may be attached to a bottom metal sheet **746**.

FIG. **7B** shows a cross sectional view of the block of flexible spacer material **700** after removal of an inner portion of the block to create an opening **745**, and the adding of vertical side foil panels **747** to opening **745**. The dimensions of opening **745** may match the dimensions of solid metal block **142**.

FIG. 7C shows a solder layer **14-1** dispensed on bottom metal sheet **746** and solid metal block **142** placed on solder layer **14-2** in opening **745**.

FIG. 7D shows a solder layer **14-2** introduced in a space between vertical side foil panels **747** and the solid metal block **142** in opening **745**, and a solder layer **14-3** dispensed on top of the assembly. Further, a top sheet of metal (e.g., metal sheet **147**) is placed over solder layer **14-3** on top of the assembly. The assembly may be heated for solder reflow and curing of solder layers **14-1**, **14-2**, and **14-3** to bond the top and bottom metal sheets **146**, **147** to the assembly components and, thus, form hybrid spacer block **140**.

In example implementations, the hybrid spacer blocks described herein than include both a solid block portion and a flexible block portion can improve the reliability of power modules. The flexible block portion of a hybrid spacer block can accommodate mechanical displacements of components induced by CTE mismatch between components and reduce die stress and solder peeling strain in the power module. The solid block portion can be used meet thermal and electrical performance requirements of the power module. Mechanical, thermal, and electrical performance of the power module can be optimized by adjusting the size, shape, and material of the solid block portion. A design of the flexible portion of the spacer (i.e., corrugated, fin, Z-shape, etc.) can also be used to tune the mechanical, thermal, and electrical performance characteristics.

Example power modules may include one or more power device dies (e.g., an IGBT and an FRD) that are coupled via the hybrid spacer blocks to DSC substrates. The hybrid spacer blocks may include flexible spacer blocks that are different (i.e. bigger or smaller) in size than the individual die size. FIG. 8 shows a diagram in which blocks representing an individual die **801**, a solid spacer block **802**, and a flexible spacer block **803** are overlaid for size comparison. Individual die **801** may have a characteristic size A, solid spacer block **802** may have a characteristic size B, and flexible spacer block **803** may have characteristic size C.

In example implementations, the die size A may be in the range of about 10 to 14 millimeters, and the flexible spacer block size C may be in the range of about 12 to 17 millimeters. Further, the hybrid spacer blocks may include solid spacer blocks that are different (e.g., smaller or equal) in size than the die size A. In example implementations, the solid spacer block size B may, for example, be in the range of about 6 to 10 millimeters for die size in the range of about 10 to 14 millimeters.

In example implementations of the hybrid spacer block (e.g., hybrid spacer block **140**) discussed above with reference to FIGS. 1 through 8, the flexible spacer block surrounds the solid block portion. In a die-spacer block combination (e.g., die-spacer block combination **150**), the flexible spacer block extends toward the die edges and can accommodate CTE mismatch stresses which may be pronounced at the die edges compared to the stresses at the die center. Each die in a multi-die power module may be coupled to a respective die-spacer block combination. FIG. 9 shows, for example, hybrid spacer **910** and hybrid spacer **920** that may be individually coupled to a first die (e.g., an IGBT) (not shown) and a second die (e.g., an FRD) (not shown) in a multi-die power module. Hybrid spacer **910** may include a solid metal block **912** surrounded by a flexible spacer block **914** coupled to the first die (e.g., an IGBT). Hybrid spacer **920** may include a solid metal block **922** surrounded by a flexible spacer block **924** coupled to the second die (e.g., an FRD).

In other implementations, hybrid spacer material (e.g., straight fin **201**, tilted fin **202**, bent fin **203**, curved fin **204**, and S-shaped fin **205**) may be incorporated in a hybrid spacer block in different geometries (e.g., other than surrounding a single solid metal block on all sides). FIGS. 10 through 12 shows examples of different geometries of hybrid spacer material used in hybrid spacer blocks.

FIG. 10 shows, for example, an arrangement of hybrid spacer blocks **1010** and **1020** that may be individually coupled to a first device die **1001** (e.g., an IGBT) and a second device die **1002** (e.g., an FRD) in a multi-die power module. First device die **1001** may include a signal pad region **1005** including signal pads **1001a** associated with the IGBT.

Hybrid spacer block **1010** may include a solid metal block **1012** and a flexible spacer block **1014**. Solid metal block **1012** may be comparable in size to device die **1001** (e.g., an IGBT). Flexible spacer block **1014** may be a rectangular strip having a width WS. Flexible spacer block **1014** as a strip may be attached to, and extend from, a side of solid metal block **1012**. Flexible spacer block **1014** may extend from the side of solid metal block **1012** to support (i.e., be bonded to) only signal pad region **1005** of first device die **1001**, while solid metal block **1012** may support the remainder of the die in a die-spacer block combination.

In the example of FIG. 10, spacer block **1020** (that forms a die-spacer block combination with second device die **1002** (e.g., an FRD)) may include only a solid metal block **1022** and no flexible spacer block.

FIG. 11 shows, for example, a hybrid spacer block **1130** that may be coupled to both first device die **1001** (e.g., an IGBT) and second device die **1002** (e.g., an FRD) in a multi-die power module. First device die **1001** may include a signal pad region **1005** and a source pad **1001b**. Second device die **1002** may include a source pad **1002b**. Hybrid spacer block **1130** may include a single solid metal block **1132** that can form a dual die-spacer block combination with first device die **1001** and second device die **1002**. Hybrid spacer block **1030** may, like hybrid spacer block **1010** (FIG. 10), further include flexible spacer block **1014** extending from a side of solid metal block **1130** to support only signal pad region **1005** of first device die **1001**.

FIG. 12 shows, for example, a hybrid spacer block **1240** that may be coupled to both first device die **1001** (e.g., an IGBT) and second device die **1002** (e.g., an FRD) in a multi-die power module. Hybrid spacer block **1240** may, like hybrid spacer blocks **1010** and **1020** (FIG. 10), include a first sub-block (i.e., a solid metal block **1212**) and a second sub-block (i.e., a solid metal block **1222**) that can be individually coupled to first device die **1001** and second device die **1002**, respectively. Hybrid spacer block **1240** may further include a single flexible spacer block **1214** that supports signal pad region **1005** of first device die **1001** and extends away continuously from signal pad region **1005** to surround solid metal block **1212** and solid metal block **1222**.

FIG. 13 shows an example method for using a flexible spacer block in a power module to accommodate component displacement that may occur, for example, due to CTE mismatch. The power module may include one or more device die (e.g., a FRD, an IGBT, etc.).

Method **1300** includes coupling an inner portion of a semiconductor device die to a solid spacer block (**1310**), disposing a flexible spacer block adjacent to the solid spacer block (**1320**), and coupling an edge portion of the semiconductor device die to the flexible spacer block (**1330**).

In method **1300**, disposing a flexible spacer block adjacent to the solid spacer block can include disposing the solid

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spacer block in an opening in the flexible spacer block. Further, disposing a flexible spacer block adjacent to the solid spacer block can, alternatively or additionally, include disposing the flexible spacer block as strip attached to, and extending from, a side of the solid spacer block.

In method **1300**, coupling the edge portion of the semiconductor device die to the flexible spacer block may include coupling the flexible spacer block to a signal pads region at the edge of the device die.

It will be understood that, in the foregoing description, when an element, such as a layer, a region, a substrate, or component is referred to as being on, connected to, electrically connected to, coupled to, or electrically coupled to another element, it may be directly on, connected or coupled to the other element, or one or more intervening elements may be present. In contrast, when an element is referred to as being directly on, directly connected to or directly coupled to another element or layer, there are no intervening elements or layers present. Although the terms directly on, directly connected to, or directly coupled to may not be used throughout the detailed description, elements that are shown as being directly on, directly connected or directly coupled can be referred to as such. The claims of the application, if any, may be amended to recite exemplary relationships described in the specification or shown in the figures.

As used in the specification and claims, a singular form may, unless definitely indicating a particular case in terms of the context, include a plural form. Spatially relative terms (e.g., over, above, upper, under, beneath, below, lower, and so forth) are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. In some implementations, the relative terms above and below can, respectively, include vertically above and vertically below. In some implementations, the term adjacent can include laterally adjacent to or horizontally adjacent to.

Some implementations may be implemented using various semiconductor processing and/or packaging techniques. Some implementations may be implemented using various types of semiconductor processing techniques associated with semiconductor substrates including, but not limited to, for example, Silicon (Si), Gallium Arsenide (GaAs), Gallium Nitride (GaN), Silicon Carbide (SiC) and/or so forth.

While certain features of the described implementations have been illustrated as described herein, many modifications, substitutions, changes and equivalents will now occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the scope of the implementations. It should be understood that they have been presented by way of example only, not limitation, and various changes in form and details may be made. Any portion of the apparatus and/or methods described herein may be combined in any combination, except mutually exclusive combinations. The implementations described herein can include various combinations and/or sub-combinations of the functions, components and/or features of the different implementations described.

What is claimed is:

1. A method, comprising:

shaping a block of flexible spacer material, the shaping including cutting an opening in a portion of the block of flexible spacer material to receive a solid metal block; and

coupling the solid metal block received in the opening to the portion of the block of flexible spacer material.

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2. The method of claim 1, wherein the block of flexible spacer material has a corrugated structure made of alternate rectangular ridges and grooves with vertical sidewalls.

3. The method of claim 1, wherein the block of flexible spacer material has a corrugated structure including flexible structures that have non-vertical sidewalls.

4. The method of claim 1 further including:

coupling a semiconductor die to the solid metal block in a semiconductor device module.

5. The method of claim 4, wherein coupling the semiconductor die includes coupling an inner portion of the semiconductor die to the solid metal block, and coupling an outer portion of the semiconductor die to the block of flexible spacer material.

6. A method, comprising:

shaping a block of flexible spacer material, the shaping including defining an opening in a portion of the block of flexible spacer material to receive a solid metal block; and

disposing the solid metal block in the opening of the portion of the block of flexible spacer material.

7. The method of claim 6, further comprising: applying a first bonding material to a first sheet of metal and placing the block of flexible spacer material including the solid metal block on the first sheet of metal.

8. The method of claim 7 wherein the first sheet of metal is a copper foil.

9. The method of claim 7, further comprising:

applying a second bonding material to a top surface of the block of flexible spacer material including the solid metal block placed on the first sheet of metal and placing a second sheet of metal over the second bonding material.

10. The method of claim 9, wherein the second sheet of metal is a copper foil.

11. The method of claim 9, wherein the first bonding material and the second bonding material are solders, and the method further includes reflowing the solders and heat curing the first bonding material and the second bonding material.

12. The method of claim 7, wherein coupling the solid metal block to the portion of the block of flexible spacer material includes adding side foil panels to the opening.

13. A method, comprising:

shaping a block made of flexible spacer material, the flexible spacer material including flexible structures that have non-vertical walls and are attached to a bottom foil;

forming an opening in the block of flexible spacer material;

disposing vertical side foil panels in the opening; and disposing a solid metal block on the bottom foil in the opening in the block of flexible spacer material.

14. The method of claim 13, wherein the bottom foil is a metal foil, and wherein the flexible spacer material includes a plurality of flexible fins attached to the metal foil.

15. The method of claim 14 further comprising dispensing a bottom solder layer on the metal foil in the opening, and disposing the solid metal block on the bottom solder layer in the opening.

16. The method of claim 15 further comprising: dispensing a side solder layer in a space between the vertical side foil panels and the solid metal block.

17. The method of claim 16 further comprising: disposing a top solder layer on the block of flexible spacer material including the solid metal block.

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18. The method of claim 17 further comprising reflowing the bottom solder layer, the top solder layer, and the side solder layer.

19. A method, comprising:

coupling a flexible spacer block and a solid metal block to 5
form a hybrid spacer block;

coupling an inner portion of a semiconductor die to the solid metal block;

coupling an outer portion of the semiconductor die to the flexible spacer block, the flexible spacer block being a 10
strip attached to, and extending from, a side of the solid metal block toward an edge of the semiconductor die;
and

bonding the hybrid spacer block to a substrate, the hybrid spacer block providing a thermal pathway for heat 15
generated in the semiconductor die to the substrate.

20. The method of claim 19, wherein the flexible spacer block has a corrugated structure made of at least one of a straight fin, a tilted fin, a bent fin, a curved fin, or an S-shaped fin. 20

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